

META-DIFFUSION: RESULTS ALIGNED WITH THE 18 SEPTEMBER EXPERIMENTAL PLAN

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ABSTRACT

This report maps every completed or conditionally required experiment to the ten modules of the 18 September 2026 plan. On controlled CIFAR-100 domain shifts, source identity matters, a reusable source-to-target map exists, the map is approximately linear, and its value is concentrated in the extreme target-scarce regime. The effect also survives FiLM conditioning. On the verified Fitzpatrick17k demographic experiment, however, correct- and wrong-disease source coordinates are statistically indistinguishable, performance is nearly invariant to the target-data budget, and the frozen disease classifier is too weak for a valid semantic conclusion. The real-demographic result therefore does not establish a reduction in target-population data requirements. Optional representation experiments do not reveal the proposed additive-basis advantage: FiLM is stronger under both Fitzpatrick task scarcity and the completed 50% CIFAR task-scarcity condition. All significance marks are based on paired 95% confidence intervals.

1 REPORTING CONVENTION

Results are paired within the same held-out semantic task, domain relation, support draw, and sampling noise whenever the design permits. Raw-performance tables contain only absolute means and 95% CIs. They are followed by comparison tables containing paired differences. For a lower-is-better distance, a reported paired advantage is

$$\Delta = d(\text{comparator}, \mathcal{D}_T) - d(\text{candidate}, \mathcal{D}_T), \quad (1)$$

so positive values favour the candidate. For higher-is-better consistency, the subtraction is reversed so that positive again favours the named candidate. Every comparison-table header explicitly states the subtraction and which method a positive value supports. A superscript * appears only on a paired difference whose 95% CI excludes zero; stars are not placed on raw values. Absolute entries are mean $\pm$ 95% CI. Sliced Wasserstein (SW), phototype-statistic distance, and energy MMD are lower-is-better; semantic consistency is higher-is-better. Frozen predictors are treated as measurement instruments with reported ceilings, not as ground truth.

2 RESULTS, MODULE BY MODULE

2.1 1. COMPLETE THE DEMOGRAPHIC SCARCITY EXPERIMENT

The full image archive was recovered from the public SFU Fitzpatrick17k-Renamed release. The release was used only as a byte source: every image was mapped to the official Fitzpatrick17k CSV and checked against its supplied `md5hash`. The published Fitzpatrick17k-C allow-list was then applied to remove known duplicates and erroneous images. Consequently, the NIH ChestX-ray14 fallback was not needed for the headline experiment.

Table 1: Fitzpatrick17k recovery, verification, and experiment allocation.

Check	Result
Images in SFU archive / official CSV rows	16,577 / 16,577
Images whose bytes match official md5hash	16,577
Rows retained by Fitzpatrick17k-C	11,394
Rows excluded by the cleaned release	5,183
Clean, verified images with Fitzpatrick type I–VI	11,098
Disease conditions represented	114
Conditions supporting $K_T = 20$ plus 25 target queries	18
Strictly eligible held-out-task pool	16 diseases
Final disease-level split	8 train / 4 validation / 4 test
Images allocated by the final split	3,712

Module conclusion. The data-access gate was cleared without using an unverified mirror: all 16,577 official rows were byte-verified, cleaning was applied before training, and four validation plus four test diseases were deliberately reserved. The use of eight training tasks is a split-design choice, not a claim that the dataset contains only eight usable diseases.

2.2 2. DEMOGRAPHIC EXPERIMENT PROTOCOL

The semantic task is skin condition; the source population is Fitzpatrick I–III and the target population is Fitzpatrick IV–VI. Validation and test episodes receive abundant source-population images, including for held-out diseases; the semantic task itself, not the availability of source examples, is held out from meta-training. The target support and real target query are disjoint.

Table 2: Headline Fitzpatrick17k protocol and model specification.

Component	Setting
Task/domain split	disease; Fitzpatrick I–III $\rightarrow$ IV–VI
Task partition	8 train / 4 validation / 4 test diseases
Source and target budgets	$M_S = 32$; $K_T \in \{1, 2, 5, 10, 20\}$
Evaluation	8 paired repeats; 96 images/cell; DDIM-50
Training	20,000 steps; AdamW, lr 2×10^{-4} ; EMA 0.999
Score model	32×32 small U-Net, base width 32, $k = 16$
Transport	linear $z_T = Az_S + b$; 272 parameters
Total trainable parameters	basis 1,046,803; FiLM 1,046,243
Test independence unit	disease mean (four test diseases)

Protocol conclusion. All required K_T values, the five prespecified controls, FiLM, disjoint target queries, and disease-level paired intervals were included. The FiLM advantage cannot be attributed to a larger model because FiLM has 560 fewer trainable parameters in this experiment.

Table 3: Raw distance to the real held-out target population across four test diseases. Entries are absolute mean $\pm$ 95% CI and lower is better; no significance testing is encoded in this raw-performance table. “Population only” is $\Delta_\gamma(0)$ and is not the strict $z = 0$ control.

Metric	Method	$K_T = 1$	$K_T = 2$	$K_T = 5$	$K_T = 10$	$K_T = 20$
Phototype	Target only	2.512 $\pm$ 0.435	2.515 $\pm$ 0.430	2.513 $\pm$ 0.431	2.515 $\pm$ 0.433	2.518 $\pm$ 0.444
	Source reuse	2.525 $\pm$ 0.442	2.525 $\pm$ 0.442	2.525 $\pm$ 0.442	2.525 $\pm$ 0.442	2.525 $\pm$ 0.442
	Population only	4.046 $\pm$ 0.311	4.046 $\pm$ 0.311	4.046 $\pm$ 0.311	4.046 $\pm$ 0.311	4.046 $\pm$ 0.311
	Wrong-task source + relation	2.503 $\pm$ 0.444	2.504 $\pm$ 0.445	2.508 $\pm$ 0.443	2.511 $\pm$ 0.441	2.513 $\pm$ 0.449
	Correct source + transport	2.501 $\pm$ 0.444	2.502 $\pm$ 0.447	2.507 $\pm$ 0.443	2.511 $\pm$ 0.438	2.508 $\pm$ 0.450
	FiLM + transport	2.297 $\pm$ 0.456	2.297 $\pm$ 0.456	2.297 $\pm$ 0.456	2.297 $\pm$ 0.455	2.297 $\pm$ 0.456
SW	Target only	0.362 $\pm$ 0.044	0.363 $\pm$ 0.044	0.362 $\pm$ 0.044	0.363 $\pm$ 0.045	0.363 $\pm$ 0.044
	Source reuse	0.364 $\pm$ 0.045	0.364 $\pm$ 0.045	0.364 $\pm$ 0.045	0.364 $\pm$ 0.045	0.364 $\pm$ 0.045
	Population only	0.287 $\pm$ 0.009	0.287 $\pm$ 0.009	0.287 $\pm$ 0.009	0.287 $\pm$ 0.009	0.287 $\pm$ 0.009
	Wrong-task source + relation	0.362 $\pm$ 0.044	0.362 $\pm$ 0.046	0.361 $\pm$ 0.046	0.362 $\pm$ 0.045	0.362 $\pm$ 0.044
	Correct source + transport	0.362 $\pm$ 0.045	0.362 $\pm$ 0.047	0.362 $\pm$ 0.045	0.362 $\pm$ 0.046	0.362 $\pm$ 0.045
	FiLM + transport	0.360 $\pm$ 0.047	0.360 $\pm$ 0.047	0.360 $\pm$ 0.047	0.360 $\pm$ 0.047	0.360 $\pm$ 0.047
Energy MMD	Target only	7.113 $\pm$ 1.804	7.163 $\pm$ 1.816	7.120 $\pm$ 1.814	7.132 $\pm$ 1.845	7.137 $\pm$ 1.817
	Source reuse	7.176 $\pm$ 1.828	7.176 $\pm$ 1.828	7.176 $\pm$ 1.828	7.176 $\pm$ 1.828	7.176 $\pm$ 1.828
	Population only	8.425 $\pm$ 1.482	8.425 $\pm$ 1.482	8.425 $\pm$ 1.482	8.425 $\pm$ 1.482	8.425 $\pm$ 1.482
	Wrong-task source + relation	7.099 $\pm$ 1.821	7.116 $\pm$ 1.841	7.087 $\pm$ 1.848	7.105 $\pm$ 1.846	7.099 $\pm$ 1.806
	Correct source + transport	7.095 $\pm$ 1.852	7.114 $\pm$ 1.887	7.110 $\pm$ 1.859	7.110 $\pm$ 1.867	7.109 $\pm$ 1.838
	FiLM + transport	6.901 $\pm$ 1.864	6.901 $\pm$ 1.862	6.901 $\pm$ 1.862	6.903 $\pm$ 1.861	6.903 $\pm$ 1.861

Distance conclusion. Correct-source transport is marginally closer than direct source reuse, but it is indistinguishable from wrong-source transport on every metric and at every K_T . Population-only obtains an artificially low pixel SW while being much worse in phototype space and energy MMD, indicating a generic low-frequency output rather than valid target-population transfer. FiLM has better absolute phototype and MMD values, but it does not restore disease-source specificity.

Table 4: Paired demographic comparisons corresponding to Table 3. Every entry is $\Delta = d(\text{comparator}) - d(\text{correct source+transport})$; positive values favour correct-source transport. A superscript * marks a 95% paired CI excluding zero.

Metric	Comparator ($\Delta = d_C - d_{\text{correct}}$; + = correct)	$K_T = 1$	$K_T = 2$	$K_T = 5$	$K_T = 10$	$K_T = 20$
Phototype	Target only	+0.0114 $\pm$ 0.0213	+0.0121 $\pm$ 0.0181	+0.0053 $\pm$ 0.0146	+0.0043 $\pm$ 0.0095	+0.0095 $\pm$ 0.0069*
	Source reuse	+0.0237 $\pm$ 0.0040*	+0.0224 $\pm$ 0.0049*	+0.0175 $\pm$ 0.0085*	+0.0140 $\pm$ 0.0127*	+0.0163 $\pm$ 0.0087*
	Population only	+1.5454 $\pm$ 0.1420*	+1.5441 $\pm$ 0.1440*	+1.5392 $\pm$ 0.1412*	+1.5357 $\pm$ 0.1374*	+1.5380 $\pm$ 0.1479*
	Wrong-task source + relation	+0.0021 $\pm$ 0.0104	+0.0017 $\pm$ 0.0061	+0.0010 $\pm$ 0.0085	-0.0001 $\pm$ 0.0065	+0.0042 $\pm$ 0.0098
	FiLM + transport	-0.2042 $\pm$ 0.0478*	-0.2058 $\pm$ 0.0507*	-0.2103 $\pm$ 0.0540*	-0.2135 $\pm$ 0.0558*	-0.2117 $\pm$ 0.0481*
SW	Target only	-0.0002 $\pm$ 0.0016	+0.0012 $\pm$ 0.0024	+0.0001 $\pm$ 0.0015	+0.0005 $\pm$ 0.0008	+0.0005 $\pm$ 0.0007
	Source reuse	+0.0023 $\pm$ 0.0016*	+0.0020 $\pm$ 0.0022	+0.0021 $\pm$ 0.0015*	+0.0022 $\pm$ 0.0017*	+0.0022 $\pm$ 0.0009*
	Population only	-0.0752 $\pm$ 0.0385*	-0.0755 $\pm$ 0.0403*	-0.0754 $\pm$ 0.0389*	-0.0754 $\pm$ 0.0396*	-0.0753 $\pm$ 0.0385*
	Wrong-task source + relation	-0.0001 $\pm$ 0.0011	-0.0000 $\pm$ 0.0013	-0.0011 $\pm$ 0.0013	-0.0001 $\pm$ 0.0008	-0.0002 $\pm$ 0.0006
	FiLM + transport	-0.0017 $\pm$ 0.0023	-0.0021 $\pm$ 0.0025	-0.0020 $\pm$ 0.0019*	-0.0018 $\pm$ 0.0015*	-0.0017 $\pm$ 0.0034
Energy MMD	Target only	+0.0184 $\pm$ 0.0488	+0.0493 $\pm$ 0.0733	+0.0097 $\pm$ 0.0552	+0.0223 $\pm$ 0.0298	+0.0274 $\pm$ 0.0243*
	Source reuse	+0.0814 $\pm$ 0.0625*	+0.0625 $\pm$ 0.0603*	+0.0666 $\pm$ 0.0658*	+0.0668 $\pm$ 0.0631*	+0.0670 $\pm$ 0.0102*
	Population only	+1.3295 $\pm$ 0.4949*	+1.3106 $\pm$ 0.5511*	+1.3147 $\pm$ 0.5009*	+1.3148 $\pm$ 0.5160*	+1.3151 $\pm$ 0.5205*
	Wrong-task source + relation	+0.0037 $\pm$ 0.0316	+0.0022 $\pm$ 0.0526	-0.0232 $\pm$ 0.0242	-0.0051 $\pm$ 0.0230	-0.0100 $\pm$ 0.0342
	FiLM + transport	-0.1936 $\pm$ 0.0267*	-0.2129 $\pm$ 0.0712*	-0.2084 $\pm$ 0.0160*	-0.2067 $\pm$ 0.0205*	-0.2063 $\pm$ 0.0729*

Source-identity conclusion. No correct-versus-wrong-source entry is significant. Correct transport does significantly beat direct source reuse on most cells, but FiLM is significantly better on phototype and MMD. The implementation uses the intended disease coordinate, yet the learned Fitzpatrick representation is effectively insensitive to which disease source coordinate is supplied. This supports coordinate/task collapse rather than a coordinate-indexing error.

Table 5: Paired target-data-equivalence comparison using the raw values in Table 3. Each entry is $\Delta = d(\text{target only at stated } K_T) - d(\text{correct transport at } K_T = 2)$; positive values favour correct transport at $K_T = 2$.

Metric (+ = correct transport at $K_T = 2$)	Target $K_T = 1$	Target $K_T = 2$	Target $K_T = 5$	Target $K_T = 10$	Target $K_T = 20$
Phototype	+0.0101 $\pm$ 0.0233	+0.0121 $\pm$ 0.0181	+0.0102 $\pm$ 0.0173	+0.0127 $\pm$ 0.0140	+0.0156 $\pm$ 0.0029*
SW	-0.0005 $\pm$ 0.0030	+0.0012 $\pm$ 0.0024	+0.0000 $\pm$ 0.0032	+0.0004 $\pm$ 0.0017	+0.0003 $\pm$ 0.0026
Energy MMD	-0.0006 $\pm$ 0.0972	+0.0493 $\pm$ 0.0733	+0.0055 $\pm$ 0.0955	+0.0180 $\pm$ 0.0495	+0.0229 $\pm$ 0.0736

Equivalence conclusion. The intervals are mostly compatible with equality, but target-only performance is itself nearly invariant to K_T . Therefore the planned statement “transport at $K_T = 2$ is equivalent to target-only at $K_T = 10$ ” would be misleading: the experiment does not show a genuine target-data-efficiency gain.

Table 6: Strict zero-coordinate audit at $K_T = 2$. The first block gives raw lower-is-better performance. In the comparison block, $\Delta = d(\text{control}) - d(\text{correct source+transport})$, so positive values favour correct-source transport. “Relation only” is $\Delta_\gamma(0)$; strict zero is literal $z = 0$.

Method or comparison ($\Delta > 0$: correct)	Phototype	SW	Energy MMD
Correct source + transport	2.5024 ± 0.4469	0.3622 ± 0.0468	7.1140 ± 1.8867
Relation only	4.0465 ± 0.3111	0.2867 ± 0.0094	8.4246 ± 1.4825
Strict $z = 0$	3.9891 ± 0.3135	0.2843 ± 0.0121	8.1995 ± 1.5442
Δ : relation only – correct	$+1.5441 \pm 0.1440^*$	$-0.0755 \pm 0.0403^*$	$+1.3106 \pm 0.5511^*$
Δ : strict $z = 0$ – correct	$+1.4867 \pm 0.1446^*$	$-0.0779 \pm 0.0367^*$	$+1.0855 \pm 0.4830^*$

Control conclusion. Relation-only and strict $z = 0$ are distinct controls, but neither provides a credible demographic solution. Their low SW conflicts with both fixed phototype distance and energy MMD, so SW alone is insufficient for this failure mode.

Table 7: Measurement-instrument audit for the demographic experiment.

Instrument	Held-out ceiling	Use in conclusion
Fixed phototype statistics	non-learned	Primary demographic distance
Frozen phenotype predictor	72.8% on test diseases (chance 50%)	Auxiliary consistency only
Frozen 113-way disease predictor	9.5% held-out (chance 0.9%)	Too weak for semantic claims
Pooled KID feature extractor	same disease predictor	Skipped; below 10% threshold

Module conclusion. The demographic protocol was completed, but its central claim failed: correct-source transport does not beat wrong-source transport and does not demonstrate reduced target-data requirements. Because this result is not clearly positive, the plan’s downstream demographic augmentation experiment was correctly not run.

2.3 3. ADD AN EXPLICIT ZERO-SHOT TARGET-TRANSFER RESULT

Table 8: CIFAR-100 zero-shot ($K_T = 0$) validation. Raw performance and paired comparisons are shown together. For SW, $\Delta = \text{SW}(\text{control}) - \text{SW}(\text{correct})$; for semantic consistency, $\Delta = C(\text{correct}) - C(\text{control})$. Thus positive values in both comparison columns favour correct-source transport.

Method	Raw SW ↓	Δ SW (positive: correct)	Raw semantic ↑	Δ semantic (positive: correct)
Correct source + transport	0.210 ± 0.020	–	0.111 ± 0.023	–
Source reuse	0.248 ± 0.018	$+0.0384 \pm 0.0113^*$	0.113 ± 0.023	-0.0017 ± 0.0090
Wrong source + relation	0.248 ± 0.021	$+0.0386 \pm 0.0141^*$	0.052 ± 0.010	$+0.0597 \pm 0.0204^*$
Relation only	0.236 ± 0.022	$+0.0260 \pm 0.0090^*$	0.051 ± 0.008	$+0.0606 \pm 0.0210^*$

CIFAR conclusion. With no target image, the correct source plus learned relation is closer to the real target distribution than all controls. It preserves source semantics at the level of direct reuse and is significantly more semantic than wrong-source and relation-only generation.

Table 9: Fitzpatrick17k zero-shot performance. The first block reports raw lower-is-better distances. In the comparison block, $\Delta = d(\text{comparator}) - d(\text{correct})$, so positive values favour correct-source transport.

Method or comparison ($\Delta > 0$: correct)	Phototype	SW	Energy MMD
Correct source + transport	2.503 ± 0.448	0.364 ± 0.046	7.143 ± 1.857
Source reuse	2.525 ± 0.442	0.364 ± 0.045	7.176 ± 1.828
Population only	4.046 ± 0.311	0.287 ± 0.009	8.425 ± 1.482
Wrong-task source + relation	2.503 ± 0.445	0.364 ± 0.045	7.139 ± 1.840
FiLM + transport	2.296 ± 0.456	0.360 ± 0.047	6.901 ± 1.862
Δ : source reuse – correct	$+0.0218 \pm 0.0065^*$	$+0.0004 \pm 0.0009$	$+0.0336 \pm 0.0332^*$
Δ : population only – correct	$+1.5435 \pm 0.1453^*$	$-0.0771 \pm 0.0392^*$	$+1.2816 \pm 0.5233^*$
Δ : wrong task – correct	-0.0003 ± 0.0067	-0.0001 ± 0.0005	-0.0037 ± 0.0205
Δ : FiLM – correct	$-0.2065 \pm 0.0510^*$	$-0.0036 \pm 0.0025^*$	$-0.2423 \pm 0.0575^*$

Module conclusion. Zero-target transfer is established on controlled CIFAR shifts, including source specificity. It is not established on Fitzpatrick17k because correct and wrong disease sources remain indistinguishable at $K_T = 0$.

2.4 4. REFRAME THE FiLM RESULT AS GENERALITY AND EFFICIENCY

Table 10: FiLM source-dependence on 180 paired CIFAR validation episodes. Each metric first reports raw performance. For SW, $\Delta = \text{SW}(\text{comparator}) - \text{SW}(\text{correct})$; for semantic consistency, $\Delta = C(\text{correct}) - C(\text{comparator})$. Positive Δ therefore favours correct-source transport.

Metric	Method or comparison	$K_T = 1$	$K_T = 5$	$K_T = 20$
SW	Correct source + transport	0.207 ± 0.022	0.205 ± 0.019	0.207 ± 0.021
	Mean source	0.246 ± 0.024	0.248 ± 0.020	0.250 ± 0.023
	Shuffled source	0.257 ± 0.025	0.259 ± 0.023	0.261 ± 0.024
	Relation only	0.241 ± 0.023	0.242 ± 0.020	0.245 ± 0.023
	Δ : mean – correct (positive: correct)	$+0.0397 \pm 0.0084^*$	$+0.0430 \pm 0.0091^*$	$+0.0430 \pm 0.0096^*$
	Δ : shuffled – correct (positive: correct)	$+0.0501 \pm 0.0154^*$	$+0.0533 \pm 0.0169^*$	$+0.0537 \pm 0.0167^*$
	Δ : relation – correct (positive: correct)	$+0.0342 \pm 0.0077^*$	$+0.0361 \pm 0.0082^*$	$+0.0380 \pm 0.0090^*$
Semantic consistency	Correct source + transport	0.125 ± 0.026	0.125 ± 0.028	0.125 ± 0.027
	Mean source	0.055 ± 0.008	0.053 ± 0.010	0.054 ± 0.009
	Shuffled source	0.053 ± 0.011	0.054 ± 0.013	0.052 ± 0.012
	Relation only	0.056 ± 0.009	0.048 ± 0.009	0.053 ± 0.009
	Δ : correct – mean (positive: correct)	$+0.0708 \pm 0.0226^*$	$+0.0717 \pm 0.0256^*$	$+0.0708 \pm 0.0235^*$
	Δ : correct – shuffled (positive: correct)	$+0.0724 \pm 0.0221^*$	$+0.0714 \pm 0.0254^*$	$+0.0729 \pm 0.0244^*$
	Δ : correct – relation (positive: correct)	$+0.0696 \pm 0.0211^*$	$+0.0773 \pm 0.0245^*$	$+0.0719 \pm 0.0217^*$

Generality conclusion. Every FiLM source-dependence comparison is significant for both SW and semantic consistency. The source-informed phenomenon is therefore not tied to the additive reverse-dynamics basis.

Table 11: Matched CIFAR conditioning comparison. SW values are correct-source transport; paired difference is FiLM minus basis, so positive favours the basis.

Quantity	Additive basis	FiLM
Coordinate-conditioning parameters	55,344	118,784
Score-model parameters	27.798M	27.917M
Encoder + transport parameters	0.334M	0.334M
Total evaluated trainable parameters	$\approx 28.132\text{M}$	$\approx 28.251\text{M}$
SW, $K_T = 1$	0.2064 ± 0.0222	0.2066 ± 0.0224
SW, $K_T = 5$	0.2074 ± 0.0202	0.2054 ± 0.0185
SW, $K_T = 20$	0.2128 ± 0.0224	0.2074 ± 0.0214
$\Delta = \text{FiLM} - \text{basis}$, $K_T = 1, 5, 20$ (positive: basis)	$+0.0002 \pm 0.0070, -0.0020 \pm 0.0076, -0.0055 \pm 0.0074$	

Module conclusion. The basis and FiLM tie on the headline SW transfer metric, while the basis uses 46.6% as many coordinate-conditioning parameters. This supports an efficiency claim, not a

generation-quality superiority claim. Matched latency, FLOPs, and peak-memory measurements were not recorded and are not inferred from parameter counts.

2.5 5. TEST WHETHER THE SIMPLE TASK-SPACE GEOMETRY IS REPRESENTATION-INDEPENDENT

Table 12: Linear versus nonlinear transport. Each block reports raw lower-is-better performance followed by a paired difference. For the basis, $\Delta = \text{SW}(\text{linear}) - \text{SW}(\text{residual})$, so positive favours the residual MLP. For FiLM, $\Delta = d(\text{nonlinear}) - d(\text{linear})$, so positive favours linear transport.

Representation	Metric or comparison	$K_T = 1$	$K_T = 5$	$K_T = 20$
Additive basis	Residual MLP SW	0.2102 ± 0.0190	0.2109 ± 0.0165	0.2138 ± 0.0193
	Linear SW	0.2124 ± 0.0211	0.2113 ± 0.0183	0.2174 ± 0.0210
	Δ : linear – residual (positive: residual)	$+0.0022 \pm 0.0049$	$+0.0004 \pm 0.0048$	$+0.0036 \pm 0.0050$
FiLM	Nonlinear SW	0.2066 ± 0.0224	0.2054 ± 0.0185	0.2074 ± 0.0214
	Linear SW	0.1973 ± 0.0206	0.1942 ± 0.0164	0.1987 ± 0.0193
	Δ : nonlinear – linear (positive: linear)	$+0.0093 \pm 0.0037^*$	$+0.0112 \pm 0.0040^*$	$+0.0087 \pm 0.0042^*$
	Nonlinear MMD	3.0686 ± 0.7686	2.9091 ± 0.5972	3.1186 ± 0.8714
	Linear MMD	2.8207 ± 0.6413	2.6281 ± 0.4690	2.9228 ± 0.7610
	Δ : nonlinear – linear (positive: linear)	$+0.2479 \pm 0.1658^*$	$+0.2810 \pm 0.1588^*$	$+0.1958 \pm 0.1640^*$
	Nonlinear statistic	0.2548 ± 0.0502	0.2564 ± 0.0455	0.2548 ± 0.0461
	Linear statistic	0.2378 ± 0.0473	0.2476 ± 0.0391	0.2502 ± 0.0435
	Δ : nonlinear – linear (positive: linear)	$+0.0170 \pm 0.0116^*$	$+0.0087 \pm 0.0126$	$+0.0046 \pm 0.0144$

Module conclusion. Linear transport is statistically equivalent to the residual MLP for the additive basis and significantly better in FiLM pixel-space SW and MMD. The representation-independent conclusion is stronger than planned: the source-target relation is adequately described by an approximately linear coordinate map, so linear transport should be the default.

2.6 6. RUN THE UNCONSTRAINED FINE-TUNING REFERENCE

Fine-tuning learning rates and step counts were selected only on validation tasks: 25 steps at 10^{-5} for $K_T = 1$, 25 steps at 10^{-4} for $K_T = 5$, and 400 steps at 10^{-5} for $K_T = 20$. These budgets were then frozen before the once-only test evaluation.

Table 13: CIFAR test-set full fine-tuning reference. Paired rows are comparator minus transport; positive favours transport.

Metric	Method/comparison ($\Delta > 0$: transport)	$K_T = 1$	$K_T = 5$	$K_T = 20$
SW	Target only	0.2027 ± 0.0139	0.1950 ± 0.0148	0.1840 ± 0.0111
	Transport	0.1839 ± 0.0122	0.1896 ± 0.0120	0.1896 ± 0.0122
	Full fine-tune	0.2675 ± 0.0193	0.2124 ± 0.0183	0.1465 ± 0.0091
	Target only – transport	$+0.0189 \pm 0.0092^*$	$+0.0054 \pm 0.0069$	-0.0056 ± 0.0067
	Full FT – transport	$+0.0836 \pm 0.0180^*$	$+0.0228 \pm 0.0202^*$	$-0.0431 \pm 0.0149^*$
Energy MMD	Target only	2.9767 ± 0.4119	2.5707 ± 0.3651	2.4511 ± 0.3003
	Transport	2.4096 ± 0.3095	2.4178 ± 0.2800	2.4747 ± 0.3384
	Full fine-tune	5.9775 ± 0.8855	3.6582 ± 0.5675	1.8834 ± 0.2059
	Target only – transport	$+0.5671 \pm 0.2705^*$	$+0.1529 \pm 0.1795$	-0.0236 ± 0.1922
	Full FT – transport	$+3.5679 \pm 0.7806^*$	$+1.2404 \pm 0.5851^*$	$-0.5913 \pm 0.4109^*$

Module conclusion. Transport is substantially safer than full fine-tuning at $K_T = 1$ and 5, where unconstrained optimisation overfits. At $K_T = 20$, full fine-tuning becomes significantly better. Thus the low-data advantage is not simply an artefact of withholding a stronger optimiser; the restricted coordinate space is beneficial under severe scarcity but becomes a limitation once enough target data are available.

2.7 7. OPTIONAL REPRESENTATION-STRENGTH EXPERIMENTS

The optional gate was satisfied only after demographic and zero-shot experiments were secured. On Fitzpatrick17k, 50% means four of eight training diseases and 25% means two of eight; validation

and test diseases are unchanged. The same experiment was then repeated on the original CIFAR image setting, where 50% retains 30 of 60 training fine classes while all 12 training superclasses remain represented.

Table 14: Fitzpatrick task scarcity at $K_T = 2$. Raw lower-is-better distances are followed by $\Delta = d(\text{basis}) - d(\text{FiLM})$; positive values in the comparison rows favour FiLM.

Training diseases	Method or comparison	Phototype	SW	Energy MMD
50% (4/8)	Additive basis	2.585 ± 0.544	0.351 ± 0.045	6.990 ± 1.976
	FiLM	2.376 ± 0.529	0.330 ± 0.041	6.217 ± 1.744
	$\Delta = \text{basis} - \text{FiLM}$ (positive: FiLM)	$+0.2083 \pm 0.0467^*$	$+0.0213 \pm 0.0046^*$	$+0.7726 \pm 0.2469^*$
25% (2/8)	Additive basis	2.627 ± 0.520	0.304 ± 0.041	6.203 ± 1.812
	FiLM	2.352 ± 0.477	0.288 ± 0.033	5.468 ± 1.450
	$\Delta = \text{basis} - \text{FiLM}$ (positive: FiLM)	$+0.2754 \pm 0.0624^*$	$+0.0159 \pm 0.0081^*$	$+0.7344 \pm 0.3830^*$

Fitzpatrick scarcity conclusion. The proposed stronger structural bias does not help the additive basis when disease tasks are scarce. FiLM is significantly better at both 50% and 25% on all three distances, while correct and wrong disease sources remain effectively tied.

Table 15: CIFAR 50% meta-training-task experiment. Raw performance is followed by paired comparison rows. For lower-is-better distances, $\Delta = d(\text{basis}) - d(\text{FiLM})$; for higher-is-better superclass consistency, $\Delta = C(\text{FiLM}) - C(\text{basis})$. Positive values therefore favour FiLM throughout.

Metric/comparison	$K_T = 1$	$K_T = 5$	$K_T = 20$
Basis SW	0.2097 ± 0.0189	0.2060 ± 0.0170	0.2141 ± 0.0171
FiLM SW	0.1912 ± 0.0185	0.1851 ± 0.0142	0.1920 ± 0.0156
$\Delta = \text{basis} - \text{FiLM}$ SW (positive: FiLM)	$+0.0185 \pm 0.0108^*$	$+0.0208 \pm 0.0114^*$	$+0.0221 \pm 0.0122^*$
Basis MMD	3.0305 ± 0.5120	3.0344 ± 0.4623	3.3888 ± 0.5750
FiLM MMD	2.6398 ± 0.4855	2.4322 ± 0.3778	2.7324 ± 0.4727
$\Delta = \text{basis} - \text{FiLM}$ MMD (positive: FiLM)	$+0.3907 \pm 0.2611^*$	$+0.6022 \pm 0.2983^*$	$+0.6564 \pm 0.3460^*$
Basis superclass consistency	0.3378 ± 0.0274	0.3394 ± 0.0263	0.3389 ± 0.0278
FiLM superclass consistency	0.3790 ± 0.0290	0.3882 ± 0.0292	0.3847 ± 0.0273
$\Delta = \text{FiLM} - \text{basis}$ consistency (positive: FiLM)	$+0.0411 \pm 0.0202^*$	$+0.0488 \pm 0.0223^*$	$+0.0458 \pm 0.0216^*$

CIFAR scarcity conclusion. The image-domain replication again contradicts an additive-basis scarcity advantage: FiLM is significantly better at all three target budgets on SW, MMD, and superclass consistency. Because this useful separation appeared, the requested 25% CIFAR extension was started; it remains [ongoing](#) and is excluded from the present conclusions.

Table 16: Fitzpatrick coordinate compression at $K_T = 2$. Raw lower-is-better distances are followed by paired compression differences $\Delta = d(k_{\text{low}}) - d(k = 16)$; positive values favour $k = 16$, whereas negative values favour the stated lower dimension.

Conditioning	k	Phototype	SW	Energy MMD
Additive basis	4	2.1825 ± 0.4918	0.3797 ± 0.0507	7.1651 ± 2.0164
Additive basis	8	2.2024 ± 0.5004	0.3786 ± 0.0517	7.1714 ± 2.0831
Additive basis	16	2.2670 ± 0.5096	0.3825 ± 0.0541	7.3803 ± 2.1970
	$\Delta: k = 4 - k = 16$ (positive: $k = 16$)	$-0.0845 \pm 0.0246^*$	-0.0028 ± 0.0036	$-0.2151 \pm 0.1818^*$
	$\Delta: k = 8 - k = 16$ (positive: $k = 16$)	$-0.0646 \pm 0.0126^*$	$-0.0038 \pm 0.0024^*$	$-0.2088 \pm 0.1205^*$
FiLM	4	2.2946 ± 0.5106	0.3703 ± 0.0495	7.2555 ± 2.0280
FiLM	8	2.0951 ± 0.4838	0.3892 ± 0.0507	7.4385 ± 2.0309
FiLM	16	2.0573 ± 0.4928	0.3853 ± 0.0526	7.3159 ± 2.0913
	$\Delta: k = 4 - k = 16$ (positive: $k = 16$)	$+0.2373 \pm 0.0434^*$	$-0.0151 \pm 0.0052^*$	-0.0604 ± 0.1669
	$\Delta: k = 8 - k = 16$ (positive: $k = 16$)	$+0.0378 \pm 0.0219^*$	$+0.0039 \pm 0.0025^*$	$+0.1226 \pm 0.0644^*$

Module conclusion. The additive basis meets the literal compression criterion because $k = 4$ and 8 are not worse than $k = 16$, but performance is almost invariant to K_T and `gain_vs_shuffled` remains approximately zero. The apparent low- k benefit is therefore better interpreted as regularisation under coordinate collapse, not efficient encoding of disease-specific information. Across both scarcity datasets, there is no evidence for an additive-basis representation-strength advantage.

2.8 8. DIAGNOSE THE ADDITIVE-BASIS ORACLE ADVANTAGE ONLY IF CHEAP

Table 17: Matched refinement-loss probe. Raw values are fractional loss descent from the stated start. The comparison column is $\Delta = \text{basis} - \text{FiLM}$: positive favours the basis for descent, but for non-monotone rise it means the basis path is rougher, not better.

Diagnostic	Raw additive basis	Raw FiLM	$\Delta = \text{basis} - \text{FiLM}$ (sign stated below)
Descent from transported coordinate	0.0004 ± 0.0001	0.0027 ± 0.0005	$-0.0024 \pm 0.0005^*$
Descent from $z = 0$	0.0313 ± 0.0051	0.1108 ± 0.0128	$-0.0796 \pm 0.0121^*$
Non-monotone rise along segment	0.0002 ± 0.0001	0.0000	$+0.0002 \pm 0.0001^*$

Module conclusion. The cheap diagnostic contradicts the proposed explanation: the basis does not descend further and its interpolation path is not smoother. The oracle advantage remains unexplained; it does not justify a new alignment loss or an architectural change.

2.9 9. FINAL METHOD SIMPLIFICATIONS

Table 18: Method decisions implied by the completed evidence.

Component	Evidence	Final decision
Target-time refinement	negligible/inconsistent gain after transport	Remove from headline method
Transport map	linear ties basis MLP and beats FiLM MLP	Use linear map by default
Conditioning	basis ties FiLM SW with fewer conditioning parameters	Keep basis as compact variant
Architecture claim	FiLM often matches or beats basis	Do not claim basis quality superiority
FFHQ-FSIG	not the viable natural demographic result	Retain only as stress test/limitation
Demographic downstream augmentation	generation result not clearly positive	Do not run under the plan’s gate

Module conclusion. The final method should be source-informed linear coordinate transport without target-time refinement. The additive basis is an efficiency-oriented implementation, while FiLM establishes architectural generality. No method change is warranted by the failed oracle diagnostic.

2.10 10. FINAL EXPERIMENTAL PACKAGE

The following compact tables collect the foundational results that were already established when the 18 September plan was written and the package status after executing the plan.

Table 19: Gaussian-mixture positive control. $\Delta_{\text{task}} = L(\bar{z}) - L(z_y)$; positive values favour the task-specific coordinate.

Task family	Own-coordinate loss	Mean-coordinate loss	Δ_{task} (+ = own coordinate)
Related, perturbation 0.10	0.4368	0.5907	$+0.1539 \pm 0.0241^*$
Related, perturbation 0.25	0.4750	0.6178	$+0.1428 \pm 0.0238^*$
Related, perturbation 0.50	0.5234	0.6770	$+0.1536 \pm 0.0331^*$
Related, perturbation 1.00	0.5937	0.7508	$+0.1571 \pm 0.0350^*$
Unrelated	0.4921	0.6453	$+0.1533 \pm 0.0392^*$

Positive-control conclusion. Across all mixture families, the correct coordinate significantly lowers denoising loss and the coordinate-loss profile is minimised at the true task coordinate. The coordinate mechanism works when the ground truth is known.

Table 20: Core CIFAR source-dependence result in pixel SW. Raw lower-is-better performance is followed by $\Delta = \text{SW}(\text{comparator}) - \text{SW}(\text{correct source+transport})$; positive values favour correct-source transport.

Method or comparison	$K_T = 1$	$K_T = 5$	$K_T = 20$
Correct source + transport	0.210 ± 0.019	0.211 ± 0.016	0.214 ± 0.019
Source reuse	0.248 ± 0.018	0.238 ± 0.016	0.222 ± 0.019
Target only	0.228 ± 0.018	0.222 ± 0.018	0.211 ± 0.021
Mean source	0.240 ± 0.021	0.245 ± 0.019	0.248 ± 0.021
Shuffled source	0.252 ± 0.021	0.256 ± 0.020	0.257 ± 0.022
Relation only	0.236 ± 0.022	0.239 ± 0.020	0.243 ± 0.023
Δ : source reuse — correct (positive: correct)	$+0.0373 \pm 0.0103^*$	$+0.0273 \pm 0.0090^*$	$+0.0077 \pm 0.0072^*$
Δ : target only — correct (positive: correct)	$+0.0181 \pm 0.0085^*$	$+0.0114 \pm 0.0072^*$	-0.0028 ± 0.0073
Δ : mean source — correct (positive: correct)	$+0.0301 \pm 0.0072^*$	$+0.0346 \pm 0.0085^*$	$+0.0338 \pm 0.0080^*$
Δ : shuffled source — correct (positive: correct)	$+0.0416 \pm 0.0125^*$	$+0.0453 \pm 0.0149^*$	$+0.0436 \pm 0.0144^*$
Δ : relation only — correct (positive: correct)	$+0.0254 \pm 0.0078^*$	$+0.0285 \pm 0.0088^*$	$+0.0295 \pm 0.0087^*$

Table 21: CIFAR target-scarcity curve in transformation-statistic distance. Raw lower-is-better performance is followed by $\Delta = d(\text{comparator}) - d(\text{correct source+transport})$; positive values favour correct-source transport.

Method or comparison	$K_T = 1$	$K_T = 2$	$K_T = 5$	$K_T = 10$	$K_T = 20$
Correct source + transport	0.299 ± 0.045	0.297 ± 0.045	0.299 ± 0.042	0.313 ± 0.041	0.293 ± 0.040
Source reuse	1.439 ± 0.106	1.212 ± 0.098	0.825 ± 0.086	0.630 ± 0.058	0.521 ± 0.050
Target only	0.360 ± 0.052	0.336 ± 0.051	0.304 ± 0.040	0.331 ± 0.053	0.301 ± 0.041
Δ : source reuse — correct (positive: correct)	$+1.140 \pm 0.108^*$	$+0.915 \pm 0.105^*$	$+0.526 \pm 0.099^*$	$+0.317 \pm 0.067^*$	$+0.229 \pm 0.059^*$
Δ : target only — correct (positive: correct)	$+0.061 \pm 0.051^*$	$+0.038 \pm 0.052$	$+0.005 \pm 0.036$	$+0.018 \pm 0.045$	$+0.008 \pm 0.036$

Table 22: CIFAR deployment-time source-budget sweep at $K_T = 1$ in pixel SW. Raw lower-is-better performance is followed by $\Delta = \text{SW}(\text{comparator}) - \text{SW}(\text{correct source+transport})$; positive values favour correct-source transport.

Method or comparison	$M_S = 16$	$M_S = 32$	$M_S = 64$	$M_S = 128$	$M_S = 256$	$M_S = 300$
Correct source + transport	0.216 ± 0.018	0.215 ± 0.019	0.214 ± 0.020	0.214 ± 0.020	0.214 ± 0.020	0.214 ± 0.020
Source reuse	0.253 ± 0.017	0.255 ± 0.017	0.252 ± 0.018	0.252 ± 0.018	0.252 ± 0.018	0.252 ± 0.018
Target only	0.230 ± 0.019	0.230 ± 0.019	0.230 ± 0.019	0.230 ± 0.019	0.230 ± 0.019	0.230 ± 0.019
Δ : source reuse — correct (positive: correct)	$+0.0377 \pm 0.0104^*$	$+0.0392 \pm 0.0100^*$	$+0.0386 \pm 0.0101^*$	$+0.0382 \pm 0.0100^*$	$+0.0380 \pm 0.0101^*$	$+0.0379 \pm 0.0101^*$
Δ : target only — correct (positive: correct)	$+0.0141 \pm 0.0084^*$	$+0.0143 \pm 0.0083^*$	$+0.0161 \pm 0.0085^*$	$+0.0161 \pm 0.0086^*$	$+0.0161 \pm 0.0094^*$	$+0.0160 \pm 0.0093^*$

Scarcity conclusion. Transport is most useful at very low target support and target-only catches up by $K_T = 20$. Increasing source evidence from $M_S = 16$ to 300 changes transport SW by only 0.002, so deployment-time source evidence is already saturated at the smallest tested budget.

Table 23: Final package status against the twelve requested items.

Required item	Status	Headline reading
1. Gaussian-mixture calibration	Complete	Correct coordinate wins in all five families
2. CIFAR source-dependence controls	Complete	Correct source beats mean, shuffled, and relation-only
3. Target-scarcity curve	Complete	Largest benefit at low K_T ; gone by $K_T = 20$
4. Source-budget saturation	Complete	Saturated by $M_S = 16$
5. Transport-structure ablation	Complete	Linear is sufficient
6. Additive basis vs FiLM	Complete	SW tie; basis uses fewer conditioning parameters
7. FiLM generality check	Complete	Source-specific effect reproduced
8. Explicit $K_T = 0$ transfer	Complete	Positive on CIFAR; negative on Fitzpatrick
9. Demographic scarcity	Complete	Central source-specificity claim fails
10. Unconstrained full fine-tuning	Complete	Worse at low K_T , better at $K_T = 20$
11. Paired 95% CIs	Complete	Used for all headline comparisons
12. Downstream demographic augmentation	not run by protocol	Gate not met: demographic generation not clearly positive

Module conclusion. Eleven of twelve package items are complete; the twelfth was explicitly conditional and was not triggered. The package supports the controlled-domain mechanism and its generality, but the requested viable real-demographic transfer result is negative rather than confirmatory.

3 CONCLUSIONS

Table 24: Direct answer to the question posed by each module of the 18 September plan.

Module	Answer
1	The full Fitzpatrick17k archive was recovered, hash-verified, cleaned, and split with held-out diseases; the data gate was passed.
2	The demographic protocol was completed, but correct-source transport equals wrong-source transport and does not reduce the target-data requirement.
3	Zero-shot source-informed transfer is positive and source-specific on CIFAR, but not on Fitzpatrick17k.
4	The principle generalises to FiLM; the additive basis offers conditioning-parameter efficiency, not superior generation quality.
5	The relation is adequately linear under both representations; linear transport is the final default.
6	Coordinate transport is preferable to full fine-tuning at $K_T \leq 5$, whereas full fine-tuning wins at $K_T = 20$.
7	Neither disease-task scarcity nor CIFAR task scarcity supports an additive-basis advantage; the completed evidence favours FiLM. Low- k Fitzpatrick results are confounded by coordinate collapse.
8	The cheap oracle diagnostic does not support the smoother/better-target-representation hypothesis, so no architectural change is justified.
9	The defensible method is linear source-informed transport without refinement; the basis is retained only as the compact implementation.
10	The final package is complete except for the conditionally gated downstream augmentation test; its final narrative stops before a real-demographic data-reduction claim.

The resulting claim is deliberately narrower than the original desired chain:

source identity matters on controlled shifts

→ a reusable, approximately linear map exists

→ it is most useful under extreme target scarcity

→ it generalises across basis and FiLM conditioning,

but the final step—reduction of real target-population data requirements—is not supported by the Fitzpatrick17k evidence. The correct scientific conclusion is therefore positive for the controlled mechanism and negative for current real-demographic generalisation.